Investigation: BAHEMA EDUCAÇÃO S.A. (CVM code 701), fiscal year 2013, as it was on 2024-12-31

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Emp. Adm. Part. - Sem Setor Principal (origin: FCA); group: non-financial.
Trading codes (2023 FCA, version 2, filed on 2023-06-26): BAHI3 (Ações Ordinárias, Bolsa, since 2018-01-26).

Document used

2013 DFP, version 1, filed on 2014-03-31; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2014-03-31).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=35966

Possible manipulation (engine v1, study 1)

Financial risk (context only)

Altman Z'' (financial risk, context only)
  Value: 7.434 (index)
  Components: X1 = 0.793; X2 = 0.162; X3 = −0.148; X4 = 2.565 (indices)
  Percentile in sector and year: 74 (n = 56)
  The company's own history: fewer than 5 previous fiscal years with a value
  - No cut-off zone: the available source does not define zones for Z''
  - Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
  Value: F = 2/9 (7 computable)
  Signals:
    - Return on assets (ROA): 0
    - Operating cash flow on assets (CFO): 1
    - Change in ROA (ΔROA): 0
    - Cash above earnings (ACCRUAL): 1
    - Change in leverage (ΔLEVER): 0
    - Change in current ratio (ΔLIQUID): 0
    - No equity issuance (EQ_OFFER): —
    - Change in gross margin (ΔMARGIN): —
    - Change in asset turnover (ΔTURN): 0
  - No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement
  - No long-term debt in either year: the change in leverage (ΔLEVER) is zero and the signal is 0 (project choice; the article only defines fell = 1, rose = 0, p. 8)

Changes

Restatement through the following year's comparative: no difference above the tolerance, or the next year's DFP did not exist yet on the date.

What was known on the date

- DFP 2010, v1: 2011-03-04
- DFP 2011, v1: 2012-03-16
- DFP 2012, v1: 2013-04-01
- FCA 2013, v1: 2013-04-01
- FCA 2013, v2: 2013-05-24
- DFP 2013, v1: 2014-03-31
- DFP 2014, v1: 2015-03-28

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
no opinion excerpt was found in the text
Auditor in the 2021 FCA: BDO RCS Auditores Independentes SS.
Auditor in the 2022 FCA: BDO RCS Auditores Independentes SS.
Auditor in the 2023 FCA: BDO RCS Auditores Independentes SS.
Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2015: Audilink & Cia. Auditores -> BKS Auditores. The outgoing firm audited 5 consecutive fiscal years (2010 to 2014), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).
- 2016: BKS Auditores -> Grant Thornton Auditores Independentes Ltda. The outgoing firm audited 1 consecutive fiscal years (2015 to 2015), by the start and end dates in the FCA.
- 2019: Grant Thornton Auditores Independentes Ltda -> KPMG Auditores Independentes. The outgoing firm audited 3 consecutive fiscal years (2016 to 2018), by the start and end dates in the FCA.
- 2020: KPMG Auditores Independentes -> BDO RCS Auditores Independentes SS. The outgoing firm audited 1 consecutive fiscal years (2019 to 2019), by the start and end dates in the FCA.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 36 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-10T19:37:54+00:00.

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.